Magpie — Sprint 13 spec: Phase 5 local floor (photo intelligence, zero cloud dependency)

Audience: Codex. **Author:** Claude. **Status: ready to implement. Locks:** the local-floor portion of Phase 5 (D-110, D-112, D-114, D-118, D-119; references D-115, D-116). **Facts:** F-17 (eBay sold-search URL). **References:** the Claude+ChatGPT convergence in `PHASE_5_SCOPE_DISCUSSION.md` , the zero-API north star, F-01/F-03/F-09 (eBay sold-comps human-fed; never warehouse/scrape eBay data), the human-in-the-loop AI principle.

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1. Goal

Ship the **zero-dependency** half of Phase 5 — value with **nothing leaving the box**, no API key, no cloud: (1) a one-click **angled eBay sold-search** panel on every item, (2) **OCR-assisted entry**, (3) **duplicate-image detection**, all flowing through a **per-field Approve / Edit / Reject review** so nothing a tool suggests becomes live data without an explicit human action. This sprint also **builds the suggestion/review UX that Sprint 14's cloud AI deep-dive will reuse** — proving the human-in-the-loop guarantee cheaply, before any cloud cost or credential exists.

2. Scope

IN (v1):

- **Angled eBay sold-search builder** on item detail (§4).
- **OCR-assisted entry** via a local adapter (§5) → staged field suggestions.
- **Duplicate-image detection** via perceptual hashing (§6).
- **Per-field staged-suggestion review** — Approve / Edit / Reject (§7).
- Reuse the dealer's-ledger design tokens; the review UX should feel like the eBay staging review.

OUT (explicitly):

- ❌ Any cloud AI / API key / network calls (OCR + hashing are **local**; sold-search only

opens URLs — it does not fetch them). That's all Sprint 14.

- ❌ AI numeric price bands (see D-111; Sprint 14 question anyway).
- ❌ **Scraping, fetching, caching, warehousing, or summarising eBay result contents** — Magpie builds the URL and opens it; the user reads results in their browser (F-09).
- ❌ Auto-merge of duplicates; auto-persist of any suggestion; filling valuation / grade / catalogue-ID / acquisition / cost / profit as authoritative.

3. Data model (additive migrations only)

- `FieldSuggestion` — a staged suggestion awaiting human action. Fields: item FK; `field` (which inventory field/specific it targets); `proposed_value`; `source` (`ocr` | `duplicate` | later `ai`); `confidence_band` (`high` | `medium` | `low` | `candidate`); `evidence` (short text — e.g. the OCR snippet/region or the matched-item ref); `status` (`pending` | `approved` | `edited` | `rejected`); resolved value + timestamps. **Nothing writes to the item from this table except on an explicit Approve/Edit action.**
- **Image fingerprint** — store a perceptual hash per photo (on the photo-asset model or a small derived record), so near-duplicate comparison is a cheap lookup. Computed at upload (and backfillable on demand).
- No model is needed for sold-search (built on demand).

4. Angled eBay sold-search builder (the highest-value, no-AI piece)

A function that takes an item's fields → a small set of **ebay.com.au** sold-search URLs, surfaced as a **"Search eBay sold"** panel on item detail (each opens in a new tab). **F-17:** base `https://www.ebay.com.au/sch/i.html?_nkw=<url-encoded query>&LH_Sold=1&LH_Complete=1`. Angles to generate (skip any that don't apply to the item/category):

- **Broad** — core keywords only.
- **Exact** — title / model / variety string.
- **By grade/condition** — where the category has a grade/condition (stamps/coins).
- **Auction-only** — add the auction format filter.

- **Buy-It-Now / fixed-price** — add the fixed-price filter.
- **Price-bounded** — `_udlo` / `_udhi` where a sensible band is known (e.g. from a current valuation).
- **Category-narrowed** — when the broad query would be noisy. Requirements: **URL-encode** the query correctly; label each link by its angle; **never fetch or store** the results (open only). Optional sort `_sop=13` / `_sop=12`. This panel is **pure local string-building** — no AI, no network from Magpie.

5. OCR-assisted entry (local, graceful)

- `OcrAdapter` **port** + a **local Tesseract implementation** (used if the binary is present) + a **deterministic fake** for CI. If Tesseract isn't installed, show a clear **"OCR unavailable"** state — never a broken screen.
- Flow: run OCR on an item's photo(s) → extract text → map likely fields (model / serial / year / denomination / material / brand where visible) → write `FieldSuggestion` rows (`source=ocr`, a confidence band, `evidence` = the recognised text). **These are staged, not applied.**
- CI must pass with **no Tesseract installed** (use the fake + fixtures); the local install is an enhancement, not a CI dependency.

6. Duplicate-image detection (local)

- On photo upload (and backfillable), compute a **perceptual hash**; store the fingerprint.
- Compare a new photo against existing fingerprints; **flag near-duplicates above a threshold** as candidates for **human review** (e.g. "this looks like LOT-00001 — same item?").
- **Never auto-merge.** Surface candidates; the human decides. (This is genuinely useful for bulk lots and avoiding re-listing.)

7. Per-field staged-suggestion review (the spine — reused by Sprint 14)

A review panel (on item detail) listing each `FieldSuggestion` with: **proposed value** · **confidence band** · **short evidence** · **source basis** (OCR / image detail / category inference) · **Approve / Edit / Reject**.

- **Approve** → writes the value to the item (provenance tagged by source, e.g. `ocr`).
- **Edit** → human edits, then it's their value (provenance reflects human edit).
- **Reject** → discarded; **the item is unchanged**.
- **Confidence bands, not fake percentages** (`high` / `medium` / `low` / `candidate`).
- **Low-confidence / candidate** suggestions sit in a separate **"low-confidence leads"** section (still useful as search leads), **not** mixed into the main list; only medium/high appear up top.
- **Hard guarantee:** nothing in `FieldSuggestion` alters the item without an explicit Approve/Edit. This must be **test-proven**.

8. Design

Reuse the **light dealer's-ledger** tokens from Sprint 12 (cool paper, ink, hairline rules, restrained antique-gold accent, tabular figures) via the `frontend-design` skill. The suggestion-review UX should feel like the proven **eBay staging review** (familiar pattern: staged rows → per-row human action). Calm, legible, no decoration that doesn't serve the task. Quality floor: responsive to phone, keyboard focus, reduced motion.

9. Acceptance criteria (closure gate)

The human-in-the-loop guarantee is the headline, not an afterthought.

- **Sold-search:** URL builder tested for **ebay.com.au + correct query encoding** + the angle/filter parameters (`LH_Sold` / `LH_Complete`, format, price); panel **opens** public sold/completed results; **no fetching, scraping, caching, or storing** of eBay result content (assert this).
- **OCR:** suggestions are **staged in `FieldSuggestion`, not written to the item**; the **OCR-unavailable** state is graceful; CI passes **with no Tesseract installed** (fake + fixtures).
- **Approve/Edit/Reject:** test-prove that **Approve writes**, **Edit writes the human value**, **Reject leaves the item unchanged**, and that **nothing persists without one of these actions**.
- **Duplicate detection:** flags near-duplicate photos for review and **never auto-merges**.
- **Zero dependency:** the whole sprint works with **no API key and no cloud**

connection.

- **Process:** backend + frontend tests green; **dual-lane Validation (SQLite + Postgres) green**; rebuild + restart the `Magpie` service; **desktop + phone screenshots**; additive migrations only; pre-migration encrypted backup before applying to the live runtime. Evidence as `SPRINT13_EVIDENCE.md`.

10. For Codex

Reasoning: High. Additive migrations only — **no destructive migration**, no cloud/network calls from Magpie (OCR + perceptual hashing run locally; sold-search only opens URLs), **no scraping/warehousing of eBay data** (F-09). No secrets are introduced this sprint (no API key yet). The per-field **no-auto-persist** guarantee is the critical correctness property — test it explicitly. This is the first of two Phase-5 sprints; it builds the review UX that Sprint 14's cloud deep-dive will reuse.